

SEQUENCES

1. Find the 25th term from the end of the arithmetic sequence 1, 4, 7, ..., 148
2. The sum of the first 5 terms of an arithmetic sequence is 75, and the first term is 5. Find the common difference.
3. What is the sum of the infinite geometric series $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$?
4. Find the 10th term of the arithmetic sequence: 2, 7, 12, 17,...
5. Find the number of terms in the arithmetic sequence 3, 7, 11, ..., 103.
6. The 3rd term of a geometric sequence is 12 and the 6th term is 96. Find the first term and the common ratio.
7. Find the sum of the arithmetic series: $2 + 5 + 8 + \dots + 53$.
8. If 5 times the 5th term of an arithmetic sequence equals 8 times its 8th term, find the first term.
9. How many two-digit numbers are divisible by both 3 and 5?
10. How many multiples of 3 are there between 100 and 500?
11. Find the value of k such that $2k$, $3k+1$, and $7k-1$ form an arithmetic sequence.
12. Determine if the sequence 1, 4, 9, 16,... is arithmetic, geometric, or neither.
13. The sum of an infinite geometric series is 10, and the first term is 5. Find the common ratio.
14. Find the value of x such that 3, x , 12 form a geometric sequence.
15. Find the value of x such that $2x-1$, $x+2$, and $3x+1$ form a geometric sequence.